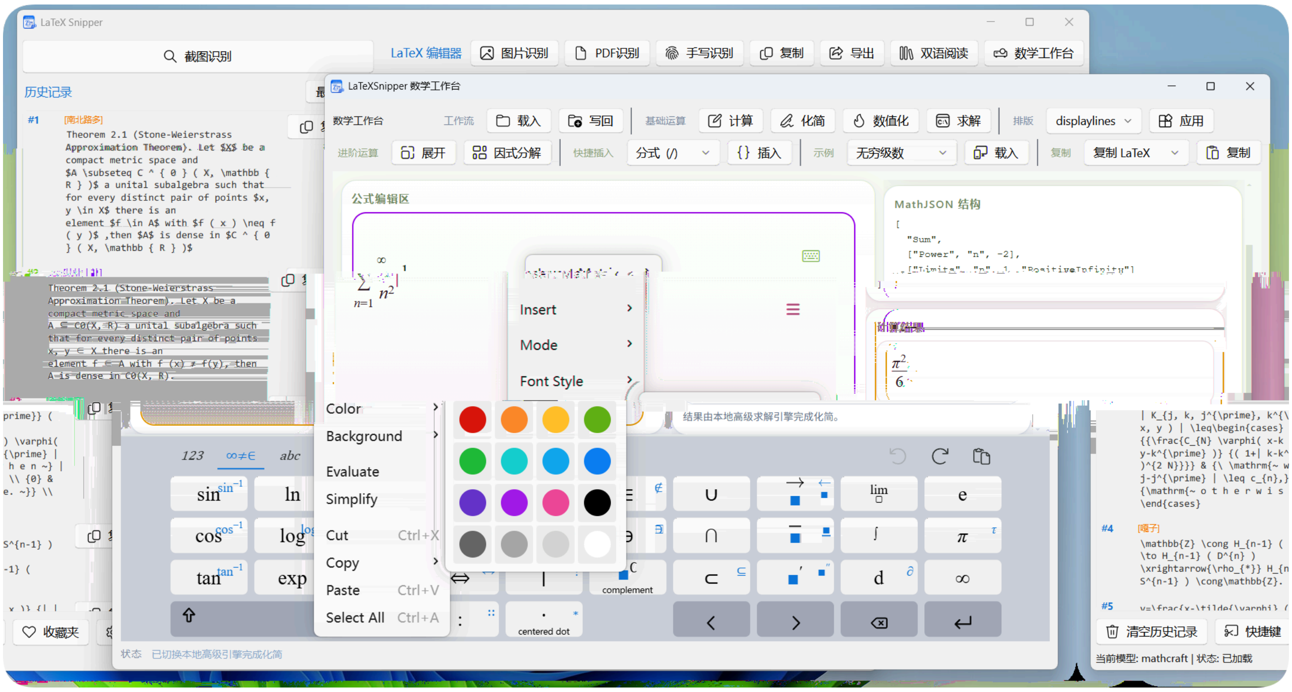
LaTeXSnipper

v2.3.2_Final_Stable |



· LaTeXSnipper

-
-
- MathCraft ONNX
-
-
-
- PDF
-
- Linux / macOS
-
-
- Bug

· MathCraft

- MathCraft OCR
-

· LaTeXSnipper

LaTeXSnipper “ ” ——
MathCraft OCR PDF/Pandoc

> ⚠️
> MathCraft PDF Pandoc
>
> Python apt/
> brew

- Windows
- Linux `python3 -m venv` Debian/Ubuntu `python3-venv`
- macOS Python 3.10+ Homebrew Python python.org

> **! 80%**
> → → **/** →
> 1.
> 2. LaTeX
> 3. LaTeX
> 4. “ ”
>
> 5. MathCraft
>
> 6. 30+ Pandoc

- **Windows** `%USERPROFILE%\.latexsnipper\logs\` `%LOCALAPPDATA%\LaTeXSnipper\logs\`
- **Linux** `~/.latexsnipper/logs/`
- **macOS** `~/.latexsnipper/logs/`
- `crash-native.log`

> ✔️
> Bug logs

LaTeX

- Ctrl+F “ ”
-
- **PDF** PDF Markdown LaTeX
-
- PDF / PDF
- MathLive
- ***/*** LaTeX Markdown MathML HTML OMML SVG Pandoc
- ***/***

“ ”

- MathCraft OCR **+**
- OpenAI-compatible Ollama MinerU GLM-OCR PaddleOCR-VL Qwen2.5/Qwen3-VL Ollama Vision MinerU Native PDF DPI OpenAI-compatible / Ollama PDF

LaTeX
MathCraft MathCraft

PDF

PDF

- MathCraft PDF
-
- MinerU Markdown Markdown LaTeX
- 5 PDF
- PDF DPI 90-300 150 DPI 200 DPI
- PDF Markdown assets

PDF OCR

- PDF PDF

- PyMuPDF PDF PDF OCR
- PDF **Poppler Fitz** Poppler Fitz
- **Argos Translate Azure Translator Google Cloud Translation DeepL API Free**
- Argos
- Azure Google DeepL

MathCraft ONNX

> ✔ **MathCraft ONNX**
>
> `warmup()` warmup
>
> `python -m mathcraft_ocr models check`

/ /

“ model xxx downloading ”

- GitHub Releases
- /
- `urllib`

- GitHub
- `HTTP_PROXY` / `HTTPS_PROXY`
-
- MathCraft
  - Windows `%APPDATA%\MathCraft\models\`
  - Linux/macOS `~/.mathcraft/models/`

> ✔
>
> `python -m mathcraft_ocr models check` `python -m mathcraft_ocr warmup --profile mixed --provider auto`

onnxruntime

`failed to import onnxruntime` `missing get_available_providers`

- `onnxruntime`
- pip
- CPU GPU

- `pip uninstall onnxruntime onnxruntime-gpu`
- GPU `pip install onnxruntime-gpu`
- CPU `pip install onnxruntime`

CUDA / GPU

warmup CUDA `cuda wasn't able to be loaded` `cublas`

- `onnxruntime-gpu` CUDA / cuDNN
- CUDA onnxruntime-gpu
-
- GPU

- **GPU** MATHCRAFT_FORCE_ORT_CPU=1 CPU
- **GPU** CUDA Toolkit cuDNN onnxruntime-gpu onnxruntime
-

/

warmup invalid protobuf failed to load model sha256 mismatch

- Windows %APPDATA%\MathCraft\models\<model_id>\ Linux/macOS ~/.mathcraft/models/<model_id>/
-
- python -m mathcraft_ocr models check

ONNX

warmup CUDA out of memory

GPU

- GPU
- MATHCRAFT_FORCE_ORT_CPU=1 CPU

“ ” “ ”

✗

_validate_config() provider mineru model_name base_url

model_name

→ Base URL “ ”

OCR 60

vLLM / Ollama / ONNX / 7B+

60

- **A** curl Ollama API
- **B** 120-180
- **C** Ollama `ollama run <model_name>`

Ollama Base URL

“ 127.0.0.1:11434 ”

- `http://localhost:11434/v1` Ollama `/v1`
- `https://` `http://` Ollama HTTPS
- Ollama
- Ollama `ollama serve`

✔

`http://127.0.0.1:11434/api/tags` JSON

/

“ : xxx ”

- Ollama `ollama list`
- OpenAI-compatible `/v1/models`

API API Key

401 “ ”

DeepSeek OpenAI API Key Ollama

API Key

⚠

API Key LaTeXSnipper API Key
Base URL

MinerU

404 409

MinerU /health /file_parse

- MinerU http://127.0.0.1:8000/health
- “ ” “ ”
- MinerU 409 MinerU

OpenAI-compatible Ollama

/v1/models 404

Ollama /api/tags /v1/models

“ OpenAI-compatible ” “ Ollama ”

!

- Ollama → **Ollama**
- / DeepSeek / OpenAI / Groq → **OpenAI-compatible**
- MinerU → **MinerU**
- MathCraft ONNX → “ ”

LaTeX
“ LaTeX ”

Python

Python

- **Windows** Python Python 3.11
- **Linux/macOS** Python 3.10+
- **/** Python 3.11 Windows

```
        创建虚拟环境

    激活

        激活

    安装公共依赖

  安装公共依赖和        快捷键依赖

  安装公共依赖
```

PyInstaller

GitHub Release exe `git clone` Windows Python 3.11 Linux/macOS PyInstaller Python

- Release
- Windows `requirements.txt` Linux/macOS `requirements-linux.txt` / `requirements-macos.txt` `requirements-build.txt`

Linux/macOS

Linux/macOS `tools/deps/python311`
MathCraft Pandoc Python

```
```

> **!**
> Linux/macOS `/usr/lib/latexsnipper` `.app` `~/.latexsnipper/deps/python311` venv `pyvenv.cfg` CI

Linux .deb python3 python3-venv macOS .deb
python3

```
或使用 官方 安装包
```

pip pywin32 / PyQt6

pip install -r requirements.txt
UI ensurepip pip setuptools / wheel /
Python Python Windows
python-3.11.0-amd64.exe Linux/macOS Python 3.10+

- pywin32 wheel
- PyQt6
- PyPI

- / PyPI

```

```

- Linux/macOS requirements-linux.txt requirements-macos.txt
- pywin32 Windows Windows pip PyPI Python wheel

Program Files /

C:\Program Files\
%APPDATA% ~/.latexsnipper/
UAC

- GitHub %LOCALAPPDATA%\LaTeXSnipper
- C:\Program Files\

/

Windows %APPDATA% ASCII C
ONNX
MATHCRAFT_HOME ASCII C:\mathcraft_data

Windows

Ollama / MinerU LaTeXSnipper
Windows Defender

- Python / Ollama
-
- Ollama `0.0.0.0` `127.0.0.1`

`api.github.com`

-
- GitHub Releases
- `%USERPROFILE%/.latexsnipper/logs/`

/

exe
PyInstaller exe

-
- **GitHub Releases** SignPath Release

PDF

PDF

PDF

- DPI < 100 > 200
- PDF
-

- PDF 140-170 DPI
- 200-300 DPI
- / “ ”
- PDF PDF Markdown/LaTeX DPI MinerU Markdown

LaTeX Markdown

PDF “ ”
MinerU

PDF PDF

/

4K

Base64 33%

-
-
-

-
- **Linux/macOS** Linux Wayland macOS
- Windows Linux macOS Dock

Pandoc

Word / EPUB / Typst
Pandoc Pandoc

- “ PANDOC ”
- Pandoc https://pandoc.org PATH
- LaTeX / Markdown / MathML / HTML Pandoc

/

- OCR MathCraft
-
-

- MathCraft
- “ ”
-

/

MathCraft /
“ ”

-
-
- OCR
  Markdown + LaTeX

PDF

- XeLaTeX `ctexart`

- amsmath amssymb mathtools geometry
- PDF
-

- *
- Compute Engine

- 2x 2*x
- SymPy/mpmath
-

/

-
- “ ” “ ” “ ”

Pandoc

.docx .epub

Pandoc UTF-8

- Pandoc ≥ 3.0
- LaTeX PDF ctex
- Word Word

PDF

PDF

- PDF OCR
- “ ”
- Argos API Key

- **PDF** OCR PDF
- Argos Translate “ Argos ”
- Azure Google DeepL “ ”

Linux / macOS

LaTeXSnipper Windows Linux macOS PDF

- Windows Win32 Linux `pynput` X11 Wayland macOS Carbon
- `Ctrl+字母` `Ctrl+Shift+字母`
- `Ctrl+F`
- Windows Qt Linux Qt `grim` `maim` `gnome-screenshot` portal macOS Qt `screencapture`
- **/** Windows “ ” Linux macOS Dock “ ”
- Windows Linux Wayland macOS Carbon
- Windows GitHub Store CPU Linux/ macOS `~/.latexsnipper/deps/python311` Python 3.10+
- Windows Inno Store MSIX GitHub Release Linux Debian/Ubuntu `.deb` macOS `.dmg` `.app.zip`

> **!**
>
> `Ctrl+字母` `Ctrl+Shift+字母`
>
> . ^

Linux: Wayland

EGL not available No physical devices GLOzone not found Failed to get system egl display

MathCraft GPU Qt WebEngine / Chromium Wayland WSL DRI render EGL/GPU

LaTeXSnipper Linux X11 + GPU + /dev/dri/renderD*

```
强制启用 图形兜底

禁用 图形兜底，尝试原生 路径
```

XWayland/XCB/QtWebEngine Debian/Ubuntu

```
```

macOS:

macOS “ ” “ ” Apple notarization

- → → “ ”
- xattr -cr /Applications/LaTeXSnipper.app

macOS:

- → → LaTeXSnipper
-

macOS:

LaTeXSnipper macOS Carbon pynput

“ ”

macOS: pip pip

pip pip / setuptools / wheel macOS Python Python 3.10+ ~/.latexsnipper/deps/python311 Python venv / ensurepip / pip

- Homebrew Python brew install python
- python.org macOS Python
- LaTeXSnipper

配置文件 日志:

依赖环境:

模型缓存 ( ):

Python

GUI

Python PATH Python

- `where python` Windows / `which python` Linux/macOS Python
-
-

/

hook

LaTeXSnipper

Bug

✗
“ ”“ ”“ ” **issue** = =

Bug

1. %USERPROFILE%\.latexsnipper\logs\ %LOCALAPPDATA%\LaTeXSnipper\logs\ **logs** stderr
2. crash-native.log
3. exe Python
4.
5.
6.

- **GitHub Issues** https://github.com/SakuraMathcraft/LaTeXSnipper/issues
- **GitHub Discussions** https://github.com/SakuraMathcraft/LaTeXSnipper/discussions

README Quick Start

README

✔

- **Windows** `python -m venv .venv` → `pip install -r requirements.txt` → `python src/main.py`
- **Linux** `python3 -m venv .venv` → `pip install -r requirements-linux.txt` → `python src/main.py`
- **macOS** `python3 -m venv .venv` → `pip install -r requirements-macos.txt` → `python src/main.py`
- `requirements-linux.txt` `requirements-macos.txt` `-r requirements.txt` Linux `pynput` macOS

`python311/` `tools/deps/python311/` Python

- `python311/` Windows Python ruff/pyright/pytest
- `tools/deps/python311/` IDE Actions `requirements*.txt` `ruff` `pytest` `pyright`
- Linux/macOS `tools/deps/python311` `~/.latexsnipper/deps/python311`

Windows

!

- `python311/` — Windows
- `tools/deps/python311/` —
- `~/.latexsnipper/deps/python311` — Linux/macOS

`docs/developer_code_standards.md`

⚠

`ruff` `pytest` `pyright` `requirements*.txt`

`pyrightconfig.json`

`tools/deps/python311` `python311/` Windows
pyright

> `tools/deps/python311` `python311`

· MathCraft

MathCraft OCR

MathCraft OCR LaTeXSnipper ONNX Runtime PDF

- GitHub https://github.com/SakuraMathcraft/MathCraft-Models
- PyPI https://pypi.org/project/mathcraft-ocr/

-
- **ONNX** ONNX Runtime CPU CUDA GPU
- PDF
-

```
或 环境：
```

PyPI ONNX Runtime `onnxruntime` `onnxruntime-gpu` LaTeXSnipper

```
检查模型缓存状态

检查运行环境并预热模型

识别图片中的公式

混合文档识别并输出
```

LaTeXSnipper “ ” MathCraft OCR + CPU GPU

`mathcraft-ocr` PyPI `0.2.x` MathCraft Models `v1.0.0` **4 ONNX**

- `mathcraft-formula-det` — Formula Detection
- `mathcraft-formula-rec` — LaTeX Formula Recognition
- `mathcraft-text-det` — Text Detection
- `mathcraft-text-rec` — Text Recognition

!

MathCraft → → + →
→

Profiles

- **`formula`** — → LaTeX
- **`text`** — OCR →
- **`mixed`** — + → Markdown

MathCraft

Abstract Algebra (p.18)

PROOF. (1) If $A \cup B = \emptyset$, then the theorem follows immediately since both A and B are the empty set. Otherwise, we must show that $(A \cup B)' \subset A' \cap B'$ and $(A \cup B)' \supset A' \cap B'$. Let $x \in (A \cup B)'$. Then $x \notin A \cup B$. So x is neither in A nor in B, by the definition of the union [redacted] so $x \notin A$ and $x \notin B$. This means $x \notin A \cup B$ and so $x \in (A \cup B)'$. Hence, $(A \cup B)' \supset A' \cap B'$ and so $(A \cup B)' = A' \cap B'$.

The proof of (2) is left as an exercise.

Example 1.4 Other relations between sets often hold true. For example,

$$(A \setminus B) \cap (B \setminus A) = \emptyset.$$

To see that this is true, observe that

$$\begin{aligned}(A \setminus B) \cap (B \setminus A) &= (A \cap B') \cap (B \cap A') \\ &= A \cap A' \cap B \cap B' \\ &= \emptyset.\end{aligned}$$

Cartesian Products and Mappings

Given sets A and B, we can define a new set $A \times B$, called the ***Cartesian product*** of A and B, as a set of ordered pairs. That is,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Example 1.5 If $A = \{x, y\}$, $B = \{1, 2, 3\}$, and $C = \emptyset$, then $A \times B$ is the set

$$\{(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)\}$$

and

$$A \times C = \emptyset.$$

We define the ***Cartesian product of*** n ***sets*** to be

$$A_1 \times \cdots \times A_n = \{(a_1, \ldots, a_n) : a_i \in A_i \text{ for } i = 1, \ldots, n\}.$$

If $A = A_1 = A_2 = \cdots = A_n$, we often write A^n for $A \times \cdots \times A$ (where A would be written n times). For example, the set $\mathbb{R}^3$ consists of all of 3-tuples of real numbers.

Subsets of $A \times B$ are called ***relations***. We will define a ***mapping*** or ***function*** $f \subset A \times B$ from a set A to a set B to be the special type of relation where each element $a \in A$ has a unique element $b \in B$ such that $(a, b) \in f$. Another way of saying this is that for every element in A, f assigns a unique element in B. We usually write $f : A \to B$ or $A \xrightarrow{f} B$. Instead of writing down ordered pairs $(a, b) \in A \times B$, we write $f(a) = b$ or $f : a \mapsto b$. The [redacted] where

$$
\begin{aligned}
&F_{11}(u_n, v_n, \delta) \\
&= -\frac{(1+b)x^*\delta}{2a} - \frac{a\sin x^* u_n^2}{2} - \frac{(1+b)x^*\delta^2}{4a^2} \\
&\quad - \frac{a\cos x^* u_n^3}{6} - \frac{(1+b)x^*\delta^3}{8a^3} + O(\|(u_n, v_n, \delta)\|^4), \\
&F_{12}(u_n, v_n, \delta) \\
&= -\frac{\left(-x^*b^2 + 2a\sin x^* - 2x^*b - x^*\right)\delta}{2a} \\
&\quad + \frac{a(b-1)\sin x^* u_n^2}{2} + \frac{(1+b)^2 x^*\delta^2}{4a^2} - \cos x^*\delta u_n \\
&\quad + \frac{a(b-1)\cos x^* u_n^3}{6} + \frac{(1+b)^2 x^*\delta^3}{8a^3} + \frac{\sin x^*\delta u_n^2}{2} \\
&\quad + O(\|(u_n, v_n, \delta)\|^4).
\end{aligned}
$$

Consider the following system

$$
\begin{pmatrix} u_{n+1} \\ v_{n+1} \\ \delta_{n+1} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_n \\ v_n \\ \delta_n \end{pmatrix} + \begin{pmatrix} \widehat{F}_{11}(u_n, v_n, \delta_n) \\ \widehat{F}_{12}(u_n, v_n, \delta_n) \\ 0 \end{pmatrix}, \tag{15}
$$

where

$$a_{11} = -1 + a\cos x^*, \quad a_{12} = 1,$$

[redacted]

then the system (15) becomes

$$
\begin{aligned}
\begin{pmatrix} \tilde{x}_{n+1} \\ \tilde{y}_{n+1} \\ \tilde{\delta}_{n+1} \end{pmatrix} &= \begin{pmatrix} -1 & 0 & 0 \\ 0 & -b + a\cos x^* & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{x}_n \\ \tilde{y}_n \\ \tilde{\delta}_n \end{pmatrix} \\
&\quad + \begin{pmatrix} \tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ \tilde{F}_{12}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ 0 \end{pmatrix},
\end{aligned} \tag{16}
$$

where

$$
\begin{aligned}
&\tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\
&= -\frac{\tan x^* \sec x^* \tilde{x}_n^2}{2a} - \frac{\tan x^* \tilde{x}_n \tilde{y}_n}{b-1} + \frac{\sin x^* \tan x^* \tilde{\delta}_n \tilde{x}_n}{2\cos x^*(a-2b-2)} \\
&\quad - \frac{a\sin x^* \tilde{y}_n^2}{2(b-1)^2} + \frac{a\sin^2 x^* \tilde{y}_n \tilde{\delta}_n}{2(b-1)(a\cos x^* - b - 1)} \\
&\quad - \frac{\tilde{\delta}_n^2}{8(a\cos x^* - b - 1)^2 a^2}\Big(\big(2x^*(b+1)^3 + a^3\sin^3 x^* \\
&\quad - 4ax^*(b+1)^2\cos x^* + 2a^2x^*(b+1)\cos^2 x^*\big)\Big) \\
&\quad + \frac{\sec^2 x^* \tilde{x}_n^3}{6a^2} + \frac{\sec x^* \tilde{x}_n^2 \tilde{y}_n}{2a(b-1)} - \frac{\tan x^* \tilde{x}_n^2 \tilde{\delta}_n}{4a(a\cos x^* - b - 1)}
\end{aligned}
$$

[redacted]

Lecture Note (p.1)

第十一届清疏大学生数学竞赛班讲义

非数学类 & 数学类共用

作者：清疏数学

组织：清疏大学

时间：July 7, 2024

版本：2.0

作者联系方式：QQ:937821418

本讲义将自动生成和 cctalk 用户绑定的 ID，如果你的 ID 多次出现在公开环境，我们会追究法律责任。

Limits and Series (p.1)

/

Problem Books in Mathematics

Ovidiu Furdui

Limits, Series, and Fractional Part Integrals

Problems in Mathematical Analysis

Springer

`block_layout_regression_v4` CUDA

- 10
- 495
- 21,417
- Markdown 304
- **8.34**
- 1.33
- 18.53

!

MathCraft OCR

- ONNX Runtime PyTorch
- manifest
- /
- OCR
-

`MATHCRAFT_FORCE_ORT_CPU` `MATHCRAFT_HOME`

```
    (永久生效):
        输入           高级   环境变量
  在 用户变量 中新建, 变量名填入上面给出的名称, 值填入   或路径

    (仅当前终端):
在              中执行:

          (永久生效):
在            或           末尾添加:

然后执行

          (仅当前终端):
```

- `MATHCRAFT_FORCE_ORT_CPU` — 1 CPU GPU
- `MATHCRAFT_HOME` — MathCraft
- `HTTP_PROXY` / `HTTPS_PROXY` —
- `MATHCRAFT_LOG_LEVEL` — DEBUG / INFO / WARNING / ERROR
- `LATEXSNIPPER_FORCE_LINUX_GRAPHICS_FALLBACKS` — Linux Qt/WebEngine
- `LATEXSNIPPER_DISABLE_LINUX_GRAPHICS_FALLBACKS` — Linux Qt/GPU
- `LATEXSNIPPER_SHOW_CONSOLE` — Windows